

Test TikZ Figures — Belief Evolution Variants

Draft visualisations for COMP3018 Set Exercises

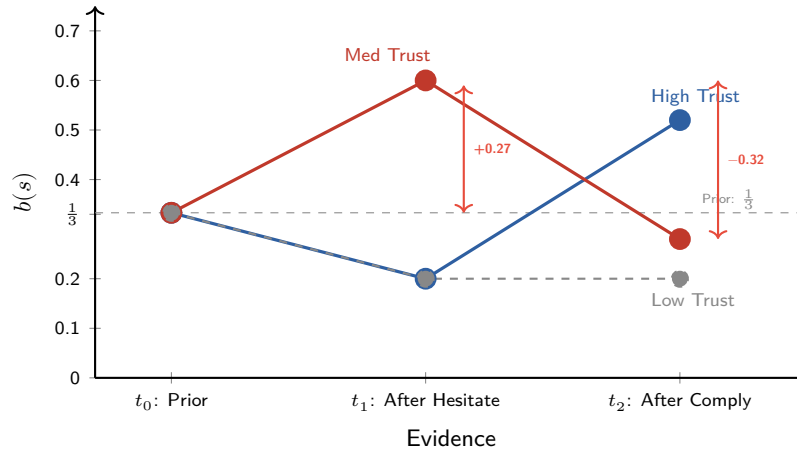


Figure 1: Simplified belief evolution (3 trust states). The medium-trust state exhibits non-monotonic behaviour: rising sharply after observing Hesitate (t_1), then falling after Comply (t_2). Red annotations show deviations from the prior, consistent with the trust recovery-despite-ongoing-faults dynamics described by Lee and See (2004) ($\sim$ non-monotonic).

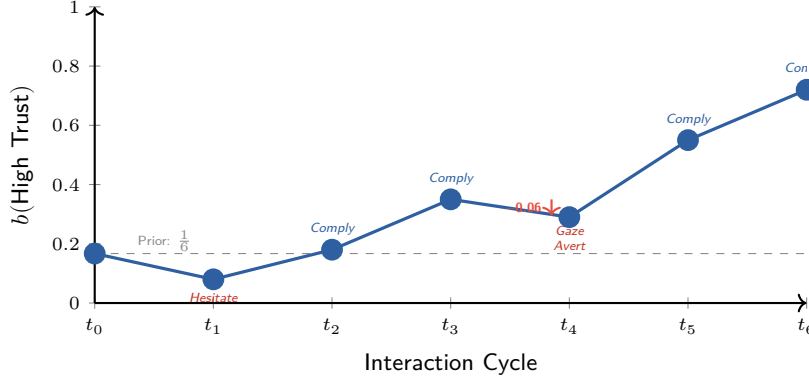


Figure 2: Trust trajectory across six interaction cycles for $b(\text{High Trust})$. Despite the overall upward trend, the posterior dips at t_1 (Hesitate) and t_4 (Gaze Avert), illustrating non-monotonic convergence: individual observations can temporarily decrease trust even as the long-run trajectory rises. Each label indicates the observation received at that timestep.

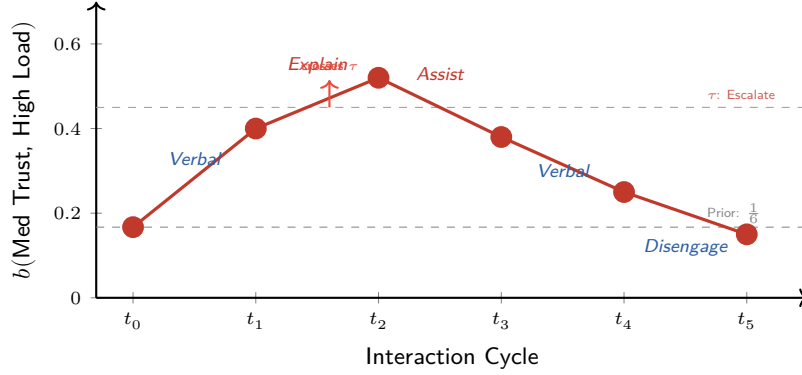


Figure 3: Action selection governed by belief thresholds. When $b(\text{Med Trust, High Load})$ exceeds the escalation threshold $\tau = 0.45$ (at t_2), the policy switches from Verbal_Remind to Explain_Benefits and then Offer_Physical_Assist. As trust recovers and the belief drops below τ (from t_3), the robot de-escalates. This demonstrates how the POMDP’s policy boundaries produce qualitatively different behaviour from a threshold-free LLM approach.

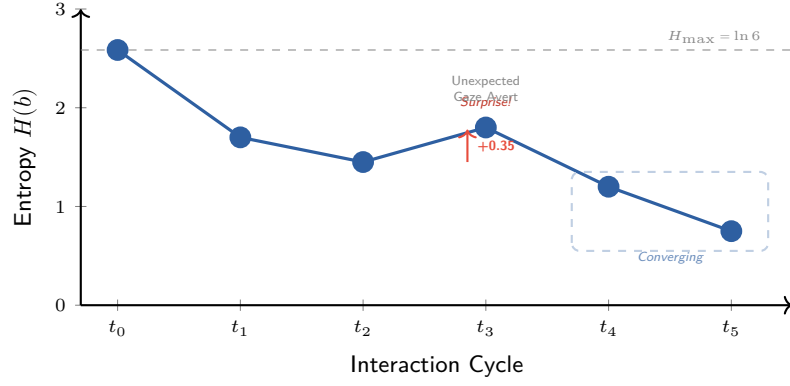


Figure 4: Entropy of the belief distribution over time. Starting from maximum entropy ($H_{\max} = \ln 6 \approx 2.58$ at the uniform prior), the robot’s uncertainty decreases as observations narrow the belief. At t_3 , an unexpected Gaze Avert *increases* entropy (+0.35), representing genuine surprise — the robot becomes less certain. This non-monotonic entropy decrease mirrors the non-monotonic posterior behaviour: evidence does not always reduce uncertainty.